Lesson Notes

MODULE 8 | LESSON 3

ESTIMATING PROBABILITY OF DEFAULT WITH BAYESIAN NETWORKS

Reading Time	30 minutes
Prior Knowledge	Stationarity
Keywords	Stationarity, Granger causality, Cointegration, Augmented Dickey-Fuller (ADF) test

In the last lesson and the lesson before, we saw how Bayesian networks can be used with time-series data (to see potential information value of lagged variables) and expert knowledge (to fully leverage the insights that come with non-quantifiable judgment and experience). In this lesson, we continue to see how Bayesian networks can meet additional estimation and prediction challenges: specifically with respect to a bank's probability of default.

1. Statistically Refreshing

We have seen Bayesian networks constructed using various metrics to describe the relationship between variables; in the last lesson, DYNOTEARS relied on mutual information, for example. In this lesson, Carraro makes use of Granger causality, with which we are also already familiar. Before starting to read this research paper, it is worthwhile to refreshing ourselves on a few of the statistical concepts that play prominent parts in the study though you may not have seen them for a while: Granger causality, stationarity, and cointegration.

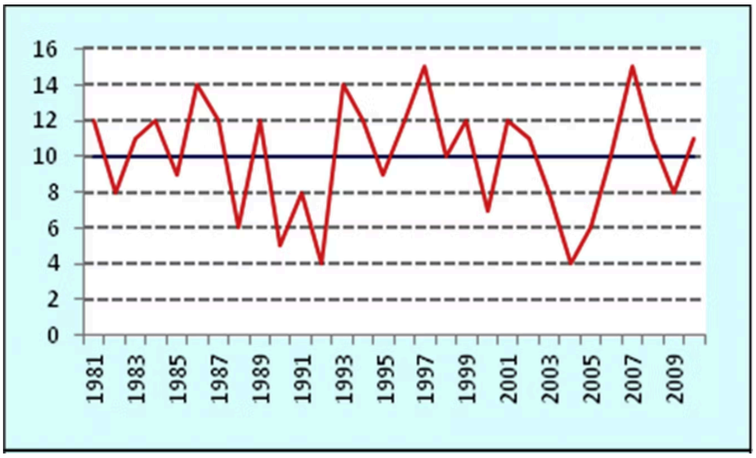
2. Granger Causality

Recall the meaning of Granger causality: X_t is said to Granger-cause Y_t if lagged elements of X_t improve the prediction of Y_t better than using only Y_t's lagged elements. As has been discussed, Granger causality is not the same as actual causality and in fact it is much closer to precedence: The behavior of X_t merely *precedes* the behavior of Y_t that X_t helps predict. A rooster's early-morning crowing noise does not cause breakfast to take place; it merely precedes it. You can also say that the crowing sound Granger-causes the breakfast meal.

3. Stationarity

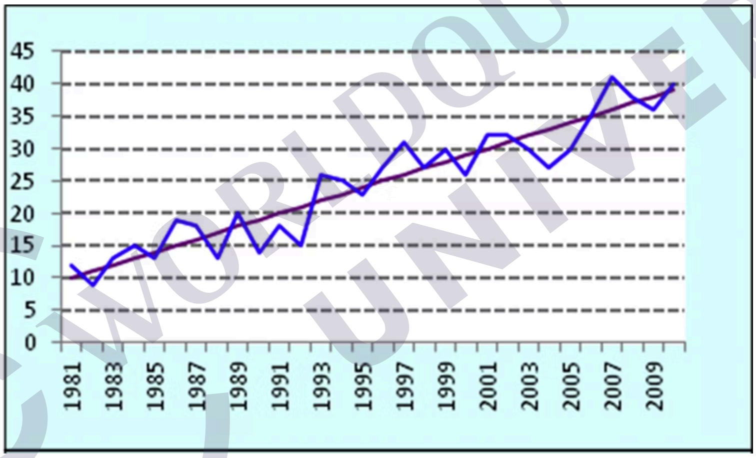
Carraro makes the important point that the investigation into Granger causality between two time series requires first of all that the two time series be stationary (90), meaning that the mean and variance of the series does not change over time—as in figure 1, as opposed to figures 2–4.

Figure 1: Stationary Time Series



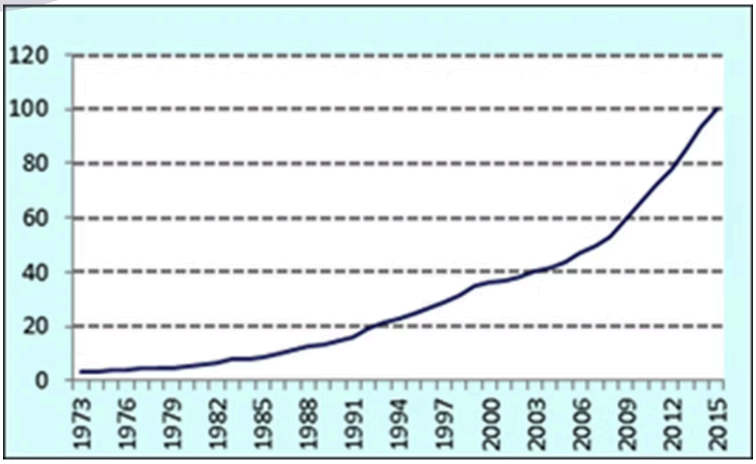
Source: Shresthaa, Min, and Guna Bhattab. "Selecting Appropriate Methodological Framework for Time Series Data Analysis." *The Journal of Finance and Data Science,* 2018, pp. 71–89, https://www.sciencedirect.com/science/article/pii/S2405918817300405?via%3Dihub.

Figure 2: Non-Stationary Time Series



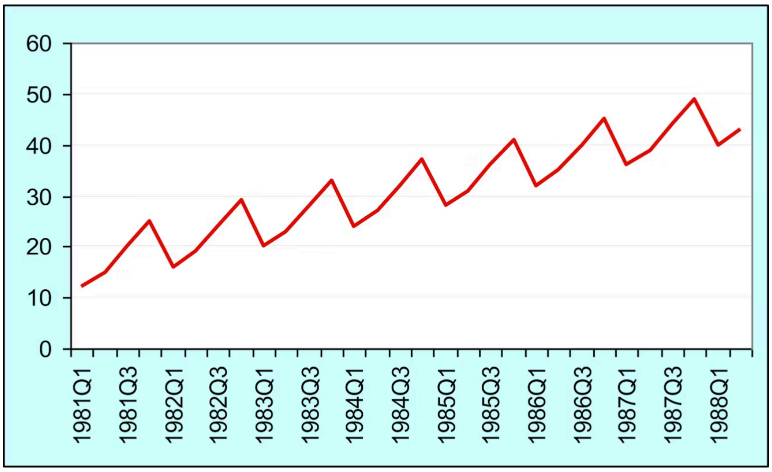
Source: Shresthaa, Min, and Guna Bhattab. "Selecting Appropriate Methodological Framework for Time Series Data Analysis." *The Journal of Finance and Data Science*, 2018, pp. 71–89, https://www.sciencedirect.com/science/article/pii/S2405918817300405?via%3Dihub.

Figure 3: Non-Stationary Time Series with Trend



Source: Shresthaa, Min, and Guna Bhattab. "Selecting Appropriate Methodological Framework for Time Series Data Analysis." _The Journal of Finance and Data Science, _2018, pp. 71–89, https://www.sciencedirect.com/science/article/pii/S2405918817300405?via%3Dihub.

Figure 4: Non-Stationary Time Series with Cycle



Source: Shresthaa, Min and Guna Bhattab. "Selecting Appropriate Methodological Framework for Time Series Data Analysis." _The Journal of Finance and Data Science, _2018, pp. 71–89, https://www.sciencedirect.com/science/article/pii/S2405918817300405?via%3Dihub.

You recall from the Financial Data (Module 6, Lesson 2) and Econometrics (Module 6, Lesson 1) courses that we test for stationarity in time series using unit root tests. The most popular of these is the Augmented Dickey-Fuller (ADF) Test, and it uses the following regression:

$$Y_t' = \phi Y_{t-1} + b_1 Y_{t-1}' + b_2 Y_{t-2}' + \ldots + b_p Y_{t-p}' \$\$$$

$$Y_t' = Y_t - Y_{t-1}$$

Remember that we need to assume that the time series is non-stationary unless we can reject the null hypothesis.

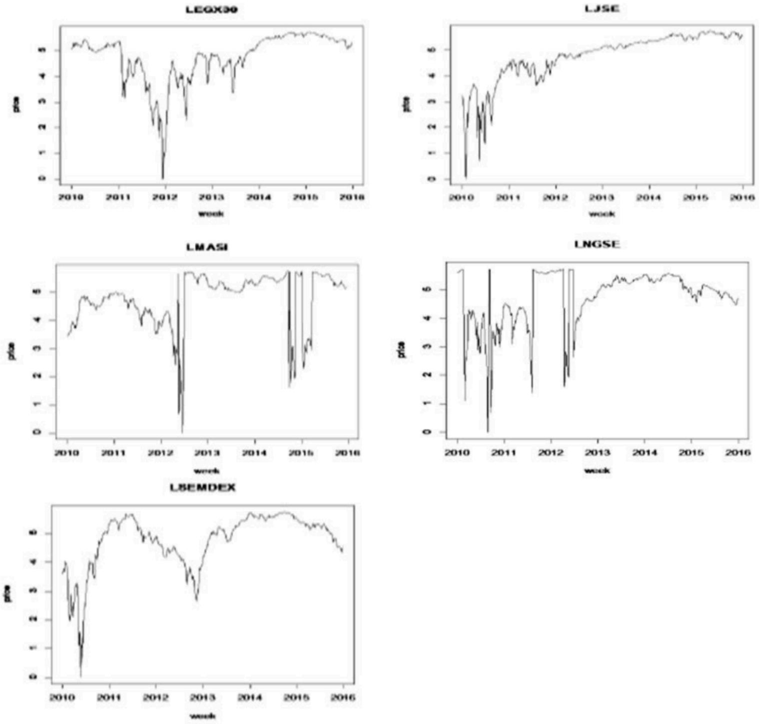
$$H_0 \ : data \ is \ nonstationary$$

$$H_A \ : data \ is \ stationary$$

4. Cointegration

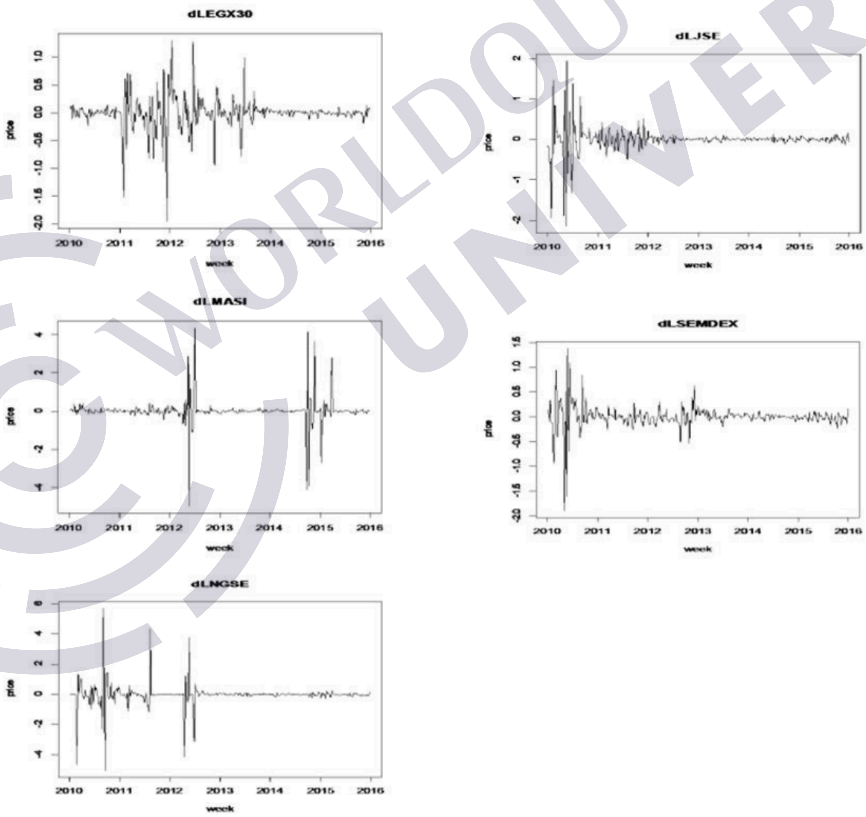
If a time series cannot be reliably considered stationary, there are methods to transform it into a stationary series, such as differencing (which can even correct for cycles like the seasonality observable in figure 4), but such transformations can degrade important long-range information that might support the discovery of Granger causality.

Figure 5: Weekly Performance of Major African Stock Indices



Source: Claver, Jimbo Henri et al. "Cointegration Analysis of Major African Stock Markets." _International Journal of Econometrics and Financial Management, _vol. 7, no. 1, 2019, pp. 37–45, http://pubs.sciepub.com/ijefm/7/1/5/index.html.

Figure 6: Weekly Performance of Major African Stock Indices



Source: Claver, Jimbo Henri et al. "Cointegration Analysis of Major African Stock Markets." *International Journal of Econometrics and Financial Management*, vol. 7, no. 1, 2019, pp. 37–45, http://pubs.sciepub.com/ijefm/7/1/5/index.html.

Take for example the weekly performance of the following African stock indices:

- Egyptian Stock Exchange (EGX 30)
- FTSE/JSE All Share Index (JSE)
- Morocco All Share Index (MASI)

- Mauritius Stock Exchange General Index (SEMDEX)
- Nigerian Stock Exchange (NGSE).

Despite their various trends and changing variances, many of them tested positive for cointegration as can be seen in figure 7, in most cases at the 1% level of significance. This may be a somewhat surprising result given that visual inspection alone may not have suggested such strong relationships.

Figure 7: Cointegration Test Results for African Stock Indices

PAIR	ADF test statistics on residuals	Results
EGX30-JSE	-2.513962	Not Cointegrated
MASI -EGX30	-4.388065***	Cointegrated
EGX30-SEM.	-2.470426	Not Cointegrated
NGSE -EGX30	-5.836424***	Cointegrated
MASI –JSE	-4.262312***	Cointegrated
JSE –SEM.	-3.024687	Not Cointegrated
JSE-NGSE	-3.48575**	Cointegrated
MASI-SEM.	-4.230202***	Cointegrated
NGSE -MASI	-5.854695***	Cointegrated
NGSE- SEM.	-6.35749***	Cointegrated

Note: *, **, *** means cointegrated at the 10%, 5% and 1% levels of significance respectively.

Source: Claver, Jimbo Henri et al. "Cointegration Analysis of Major African Stock Markets." *International Journal of Econometrics and Financial Management*, vol. 7, no. 1, 2019, pp. 37–45, http://pubs.sciepub.com/ijefm/7/1/5/index.html.

This discussion of cointegration among time series whose plots do not suggest it just points to the importance of thorough initial analysis and requisite cleaning of any time-series data. Such preliminary work should take into account the results of all relevant tests. *Carraro* provides a good review of these tests and a solid treatment of cointegration in relation to Granger causality, especially in Appendix A (89–92). A similarly scrupulous approach to handling time-series data will be outlined in detail in the reading for the next lesson due to its criticality for reliable results.

5. Conclusion

In this module, we have been exploring the power of Bayesian networks for probabilistic estimation problems, as in this lesson with the inference of a bank's probability of default. We continue this Bayesian network journey with crude oil price prediction in the next lesson.

References

- Carraro, Andrea. *Bayesian Networks and Financial Stress Testing*. 2018. Ca' Foscari University of Venice, Master's thesis, 2018, http://dspace.unive.it/handle/10579/14116.
- Claver, Jimbo Henri, et al. "Cointegration Analysis of Major African Stock Markets." *International Journal of Econometrics and Financial Management*, vol. 7, no. 1, 2019, pp. 37–45, http://pubs.sciepub.com/ijefm/7/1/5/index.html.
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